MBS-raw: Physical Optics for Ice Crystals

Manual

2026-03-22

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1 Overview

MBS-raw computes light scattering by non-spherical particles using the Physical Optics (PO) method. It traces geometric optics rays through the particle (reflections, refractions), then applies Kirchhoff diffraction to each output beam to obtain the far-field Mueller matrix.

Optimized version: ∼70× faster than original. AVX-512/AVX2 SIMD, OpenMP parallel, Sobol quasi-random orientations. Builds for Intel, AMD Zen 2 (EPYC 7H12), and AMD Zen 4 (Ryzen 7000 / EPYC Genoa).

2 Quick Start

```
# Simple run: hex column, Sobol orientations, auto theta grid
mbs_po --po --sobol 1024 --auto_tgrid \
    -p 1 10 10 -w 0.532 --ri 1.31 0 -n 12 \
    --grid 0 180 48 180 --close

# Fully adaptive (auto everything)
mbs_po --po --adaptive 0.01 --auto_tgrid \
    -p 1 10 10 -w 0.532 --ri 1.31 0 -n 12 \
    --grid 0 180 48 180 --close
```

3 CLI Reference

All flags are grouped by function. Required flags have no default; optional flags are marked with their defaults.

3.1 Particle Definition

3.1.1 `-p TYPE HEIGHT DIAMETER [EXTRA]`

Type	Shape	Parameters
1	Hexagonal column	H = height, D = diameter
2	Bullet	H, D (cap angle 62°, auto)
3	Bullet rosette	H, D, [cap height]
4	Droxtal	arg3 = sup parameter
10	Concave hexagonal	H, D, concavity depth
12	Hexagonal aggregate	H, D, number of elements

Symmetry: auto-detected. Hex column: $\beta_{sym} = \pi/2$, $\gamma_{sym} = \pi/3 \rightarrow 12\times$ reduction.

3.1.2 `-pf FILENAME`

Load particle geometry from file. `-p` argument provides the filename.

3.1.3 `-rs SIZE`

Resize particle loaded with `-pf` to maximal dimension SIZE (μm).

3.1.4 `-forced_convex` / `-forced_nonconvex`

Force convex or non-convex tracing algorithm.

3.2 Physical Parameters

3.2.1 `-w WAVELENGTH_UM`

Wavelength in micrometers. Examples: `-w 0.532` (green), `-w 1.064` (NIR).

3.2.2 `-ri N_REAL N_IMAG`

Complex refractive index $m = n_{re} + i\, n_{im}$.
Important: Without `-abs`, imaginary part is forced to zero.

3.2.3 `-abs`

Enable absorption. Requires `-w`. Uses `DiffractInclineAbs`.

3.2.4 `-n MAX_REFLECTIONS`

Value	Use case	Notes
5–6	Forward scattering, C_{sca}	$< 1\%$ error
12	General purpose	$< 5\%$ backscatter error
20	Backscattering	$< 3\%$ error
25	Reference calculations	

3.3 Method

- `-po` — Physical Optics (Kirchhoff diffraction). **Required for PO.**
- `-go` — Geometrical Optics (no diffraction).
- `-incoh` — Incoherent summation ($|J|^2$ per beam, then sum).
- `-karczewski` — Karczewski polarization matrix (experimental, same M_{11}).
- `-jones` — Output Jones matrices to file.
- `-all` — Compute all trajectory groups.

3.4 Orientations

3.4.1 `-sobol N` (recommended)

N Sobol quasi-random orientations (power of 2). Auto-uses particle symmetry.
Convergence: $O((\log N)^2/N)$, much faster than Monte Carlo $O(1/\sqrt{N})$.

x	C_{sca} ($< 2\%$)	$M_{11}(180)$ ($< 5\%$)
10–50	128	512
50–200	256	1024
200–1000	512	4096
> 1000	1024	8192+

3.4.2 `-adaptive EPS`

Auto-determine orientations. Starts at 256, doubles until C_{sca} converges to relative accuracy EPS.

3.4.3 `-sym B G`

Override particle symmetry: $\beta \in [0, \pi/B]$, $\gamma \in [0, 2\pi/G]$.
Example: `-sym 6 2` $\rightarrow \beta \in [0, 30]$, $\gamma \in [0, 180]$ ($12\times$ reduction).

3.4.4 Other orientation modes

- `-random N_BETA N_GAMMA` — Uniform grid.
- `-montecarlo N` — Random orientations.
- `-fixed BETA GAMMA` — Single orientation (degrees).
- `-orientfile FILE` — Read (β, γ) pairs (radians) from file.
- `-b MIN MAX`, `-g MIN MAX` — Override ranges (degrees).

3.5 Scattering Grid

3.5.1 `-grid T1 T2 N_PHI N_THETA`

Uniform grid. **Use** $N_\phi \geq 48$; $N_\phi = 6$ causes 50–200% error.

3.5.2 `-auto_tgrid` (recommended)

Auto non-uniform θ grid with 5 adaptive zones:

Zone	Range	Step	Purpose
Fine	0 to 10×peak	peak/20	Forward peak
Transition	to 20	geometric ×1.3	Smooth bridging
Medium	to 120	0.5 (x<100) or 1	Side scattering
Side/back	to 175	1	Backscattering
Near-backscatter	to 180	0.25	LDR

[width=]fig$_a uto_g rid_p oints.pdf$

Figure 1: Left: total N_θ and N_ϕ vs size parameter x. Right: zone breakdown of θ-grid points.

[width=0.85]fig$_a uto_g rid_s teps.pdf$

Figure 2: Step size $\Delta\theta$ vs scattering angle for different x. Small x has coarser steps (broad peak); large x needs fine steps near θ=0.

3.5.3 `-auto_phi`

Auto-select N_ϕ based on x: $N_\phi = \max(48,\ 6\lceil\sqrt{x}/1.5\rceil)$.

[width=]fig$_a uto_p hi.pdf$

Figure 3: Left: N_ϕ from auto_phi (blue) vs fixed values. Right: Q_{sca} error vs N_ϕ for different sizes—larger x needs more ϕ bins.

Recommended: use both `-auto_tgrid -auto_phi` together.

3.5.4 `-tgrid FILENAME`

Custom θ grid from file (degrees, one per line).

3.6 Multi-Size (`-sizefile`)

```
--sizefile sizes.txt   # one x = pi*D/lambda per line
```

Traces beams once, caches geometry, recomputes diffraction for each size. Output: `M_x10.dat`, `M_x20.dat`, etc.

3.7 Performance Options

- `-beam_cutoff EPS` — Skip negligible beams.
- `-shadow_off` — Disable shadow beam (debug).
- `-r RATIO` — Small beam restriction ratio (default 100).

3.8 Output

- `-o DIRNAME` — Output folder (default `M`).
- `-close` — Exit after calculation (**required for scripts**).
- `-log SECONDS` — Progress output interval.
- `-tr FILE` — Trajectory groups from file.
- `-gr` — Per-group output.

4 Output Files

4.1 M.dat

Tab-separated columns:

```
theta  2pi*dcos  M11  M12  M13  M14  M21  M22  M23  M24  ...  M44
```

$C_{sca} = \sum M_{11} \times (2\pi\, d \cos\theta)$.

5 Build

5.1 Intel / AVX-512

```
bash build.sh            # -> bin/mbs_po
```

`-O3 -march=native -mavx512f -mavx512dq -fopenmp`. Requires GCC ≥ 9.

5.2 AMD EPYC 7H12 (Zen 2, SP3)

```
bash build_epyc.sh       # -> bin/mbs_po_epyc
bash build_epyc_clang.sh # -> bin/mbs_po_epyc_clang
```

`-march=znver2`. No AVX-512; `fast_sincos_8x` falls back to 2×`fast_sincos_4x`.

NUMA: `OMP_PROC_BIND=close OMP_PLACES=cores OMP_NUM_THREADS=64`

5.3 AMD Zen 4 (Ryzen 7000, EPYC Genoa)

```
bash build_zen4.sh       # -> bin/mbs_po_zen4
```

`-march=znver4 -mavx512f -mavx512dq -mavx512vl`.

5.4 Generic AVX2

Remove `-mavx512*` from `build.sh`. Code auto-detects via `#ifdef __AVX512F__`.

6 Performance

6.1 Single-thread benchmarks

Hex D=H=10 μm, 128 Sobol, 48 $\phi \times$ 181 θ, n=6:

Version	Phase 2	vs Original
Original (before optimizations)	∼320 s	1×
Optimized + batched sincos	4.6 s	∼**70**×
EPYC build (AVX2 only)	4.7 s	∼68×

6.2 Key optimizations

1. **PrecomputeEdgeData**: slopes, intercepts computed once per beam (not per direction).
2. **Theta-coefficients**: $A(\theta) = \sin\theta \cdot a_s + \cos\theta \cdot a_c + a_0$ (2 FMA).
3. **Batched sincos**: all vertex phases for all θ pre-computed before the θ-loop. AVX-512 `fast_sincos_8x` or AVX2 2×`fast_sincos_4x`.
4. **Polynomial sincos**: Cody–Waite range reduction + Chebyshev polynomial (∼14 decimal digits).
5. **Inline hot loop**: no heap allocation, stack-only `complex` arithmetic.
6. **OpenMP**: thread-local Mueller accumulators, `schedule(dynamic,1)`.
7. **Sobol + symmetry**: native C++ Sobol generator with particle symmetry reduction.

6.3 OpenMP scaling

Threads	Phase 2	Speedup
1	4.6 s	1.0×
4	1.7 s	2.7×
12 (SMT)	∼0.9 s	∼5×
64 (EPYC)	∼0.2 s*	∼23×*

*Estimated; use ≥512 orientations for 64 cores.

7 Bugfixes

7.1 Forward-direction Fresnel sign (2026-03-22)

Bug: sign error in the forward-direction special case (A≈0, B≈0) of the Kirchhoff integral. The edge-sum converges to $-F_{inv} \cdot area$ but the shortcut used $+F_{inv} \cdot area$.

Symptom: $M_{11}(0)$ anomalously small in coherent mode for small particles. Example (hex D=H=3 μm): $M_{11}(0)/M_{11}(0.1) = 0.019$ (should be ≈ 1.03).

Fix: negate the forward-direction formula in 6 locations:

- `HandlerPO.cpp`: lines 970, 1333, 1889
- `Handler.cpp`: lines 255, 473, 642

Incoherent mode unaffected ($|J|^2$ is sign-independent). Q_{sca} changed by ∼0.06%. See `BUGFIX_forward_direct` for full details.

8 Known Limitations

1. $Q_{sca} > 2$ **at large** x: PO does not enforce the optical theorem.
2. M_{33}, M_{34}, M_{44}: RotateJones vs Karczewski give different off-diagonal elements (same M_{11}).
3. **Small particles** ($x < 20$): PO not valid. Use ADDA or T-matrix.
4. **Backscattering convergence**: $M_{11}(180)$ needs 10–100× more orientations.
5. **LTO**: `-flto` causes ∼20% regression. Do not use.

9 File Listing

File	Description
`build.sh`	Intel/AVX-512 build
`build_epyc.sh`	AMD EPYC 7H12 (Zen 2) build
`build_epyc_clang.sh`	EPYC + Clang/AOCC build
`build_zen4.sh`	AMD Zen 4 (Ryzen 7000/Genoa) build
`Makefile`	Legacy Makefile (Intel)
`Makefile.epyc`	Makefile for EPYC
`tests/run_tests.sh`	Regression test suite
`MANUAL.md`	This manual (Markdown source)
`BUGFIX_forward_direction_sign.md`	Bugfix documentation